
Comparative Evaluation of AI-Guided Rigid Linker Design for Bispecific T-Cell Engagers: Balancing Geometric Precision and Structural Confidence

Anonymous Author(s)

Abstract

Bispecific T-cell engagers (BiTEs) require precise spatial control to bridge the immunological synapse, with an optimal inter-membrane distance of approximately 130 Å. Current designs employ flexible (G₄S)_n linkers susceptible to entropic collapse. We systematically compared three computational approaches for designing rigid linkers in an anti-HER2 × anti-CD3 BiTE: unconstrained de novo generation, geometry-guided inpainting, and phase-aligned de novo design (PADD). De novo generation failed completely (mean RMSD 51.4 Å, minimum 36 Å). Initial screening revealed a trade-off: inpainting achieved superior geometric precision (distance error 49.7 vs. 96.3 Å, $p < 0.001$), while PADD showed higher AlphaFold2 confidence (linker pLDDT 73.4 vs. 71.2, $p = 0.014$). However, extended validation under stress conditions revealed that inpainting candidates exhibited superior reproducibility (distance SD 29.5 Å vs. 62–70 Å for PADD) and achieved the 130 Å target precisely (mean 130.8 Å). These findings suggest that AlphaFold2 confidence metrics may reflect familiarity with canonical structures rather than genuine stability, and that granting AI partial geometric freedom within human-defined constraints produces more robust designs than strict human specification of geometry.

1 Introduction

Bispecific T-cell engagers (BiTEs) represent a transformative class of cancer immunotherapeutics that redirect cytotoxic T-cells to eliminate tumor cells by simultaneously binding CD3 on T-cells and tumor-associated antigens (TAAs) [1, 2]. The therapeutic mechanism fundamentally depends on forming a productive immunological synapse (IS)—a specialized cell-cell interface enabling directed cytotoxic granule release and subsequent tumor cell lysis [3, 4]. While BiTEs have achieved remarkable clinical success in hematologic malignancies, their translation to solid tumors remains challenging due to dose-limiting cytokine release syndrome (CRS), on-target off-tumor toxicity, and insufficient efficacy [5–7]. A 2024 systematic review established a direct exposure-response relationship between maximum drug concentration and CRS incidence [8], underscoring the urgent need to redesign BiTE molecular architecture.

Central to BiTE function is the linker domain connecting the two antigen-binding scFv fragments. Conventional designs employ flexible glycine-serine (G₄S)_n linkers that facilitate protein folding but suffer from a critical biophysical liability: entropic collapse [9, 10]. In solution, flexible linkers spontaneously adopt compact conformations that minimize their end-to-end distance, effectively shortening the BiTE below the threshold required for productive synapse formation [9, 11]. According to the kinetic segregation model of T-cell activation, effective engagement requires excluding the inhibitory phosphatase CD45 from the synaptic cleft, which necessitates precise control of inter-membrane distance [3, 12]. Foundational studies by Santich *et al.* demonstrated that interdomain spacing and spatial configuration directly drive the potency of T-cell bispecific antibodies [12]. Recent work by Leithner *et al.* using SAXS confirmed that forming a close-contact synapse of approximately 130 Å is a critical determinant of BiTE potency, and crucially identified molecular flexibility as an independent negative predictor of efficacy—less flexible T-cell engagers demonstrated superior signaling and cytotoxicity even when comparing constructs with similar synapse distances [3]. Systematic geometric engineering by Zhang *et al.* further established that T-cell killing progressively diminishes as the distance between CD3 and TAA binding domains increases [15], while Zhou *et al.* showed that cyclized variants with constrained linker geometry exhibit superior cell-killing efficacy and improved thermodynamic stability compared to linear counterparts [1]. These converging findings establish that both rigidity and precise spatial control—not merely approximate distance—are essential design parameters for next-generation BiTEs.

Traditional rigid linker motifs, such as α -helical (EAAAK)_n repeats, offer improved structural definition

compared to flexible sequences [9, 13], but introduce a critical challenge: helical phase alignment. Because α -helices rotate approximately 100° per residue around their axis, the terminal orientation is exquisitely sensitive to length—a single residue addition rotates the exit vector by nearly one-third of a full turn [13, 14]. Without precise control of this rotational phase, helical linkers can orient the scFv domains into non-productive configurations or introduce angular strain at domain junctions [15, 16]. Meanwhile, empirical library screening is fundamentally limited by its inability to simultaneously satisfy the multiple geometric constraints required for optimal synapse formation [3, 17].

The emergence of generative AI tools for protein design, particularly RFdiffusion [18], has opened new possibilities for solving such complex constraint satisfaction problems. However, unconstrained de novo generation fails to spontaneously discover the precise geometry required for BiTE function, as these models optimize for general protein stability rather than specific spatial requirements [18, 19]. Recent studies demonstrate that RFdiffusion excels at motif scaffolding and inpainting tasks when provided with appropriate geometric constraints [18, 20], while ProteinMPNN can optimize amino acid composition for fixed backbone geometries [21]. This suggests that successful AI-assisted BiTE design may require strategically partitioning the design problem between human expertise and AI capability.

We systematically compare two distinct modalities of human-AI collaboration for designing rigid BiTE linkers. In *Geometry-Guided Inpainting*, human designers define the structural constraints while RFdiffusion solves the geometric problem of connecting endpoints, leveraging the ability of diffusion models to discover non-intuitive structural solutions including unexpected bends or turns [18, 20, 22]. In *Phase-Aligned De Novo Design (PADD)*, human designers calculate precise helical phase alignment and fix complete backbone geometry, while ProteinMPNN optimizes only the sequence [21]. We evaluate both against unconstrained de novo generation using AlphaFold2 validation [23], Normal Mode Analysis (NMA), and inter-domain distance measurements. Our target is an anti-HER2 $\times$ anti-CD3 BiTE achieving 130 Å inter-antigen distance with minimal flexibility.

2 Materials and methods

2.1 Construction of reference coordinates

The BiTE model comprised two scFvs targeting HER2 and CD3, respectively. The anti-HER2 scFv was constructed using V_H and V_L domain coordinates derived from trastuzumab (PDB: 1N8Z), while the anti-CD3 scFv was constructed from OKT3 (PDB: 1SY6). Each scFv was connected via a canonical $(G_4S)_3$ linker. Antigen-scFv complexes were assembled by structurally aligning the CD3 ϵ extracellular domain (PDB: 1XIW) and a full-length AlphaFold2 model of HER2 to their respective scFvs. Based on the kinetic segregation model [3, 12], we established a "130 Å Reference State" as the ground truth geometry, defined by setting the inter-membrane distance—the Euclidean distance between membrane-proximal anchor points—to exactly 130 Å (Figure 1a,b).

2.2 Unconstrained de novo generation (negative control)

To establish a baseline for AI-only design without human geometric guidance, RFdiffusion (v1.1.1) was used to generate de novo linkers spanning the gap between the two scFvs (Figure 1c). The linker region was allowed to vary in length from 50 to 70 residues in 5-residue increments. No secondary structure constraints were applied, granting the model complete freedom to determine both backbone geometry and topology. For each length group, 8 independent backbone trajectories were generated with 4 structural variants per trajectory, yielding 160 total designs evaluated solely by RMSD to the Reference State.

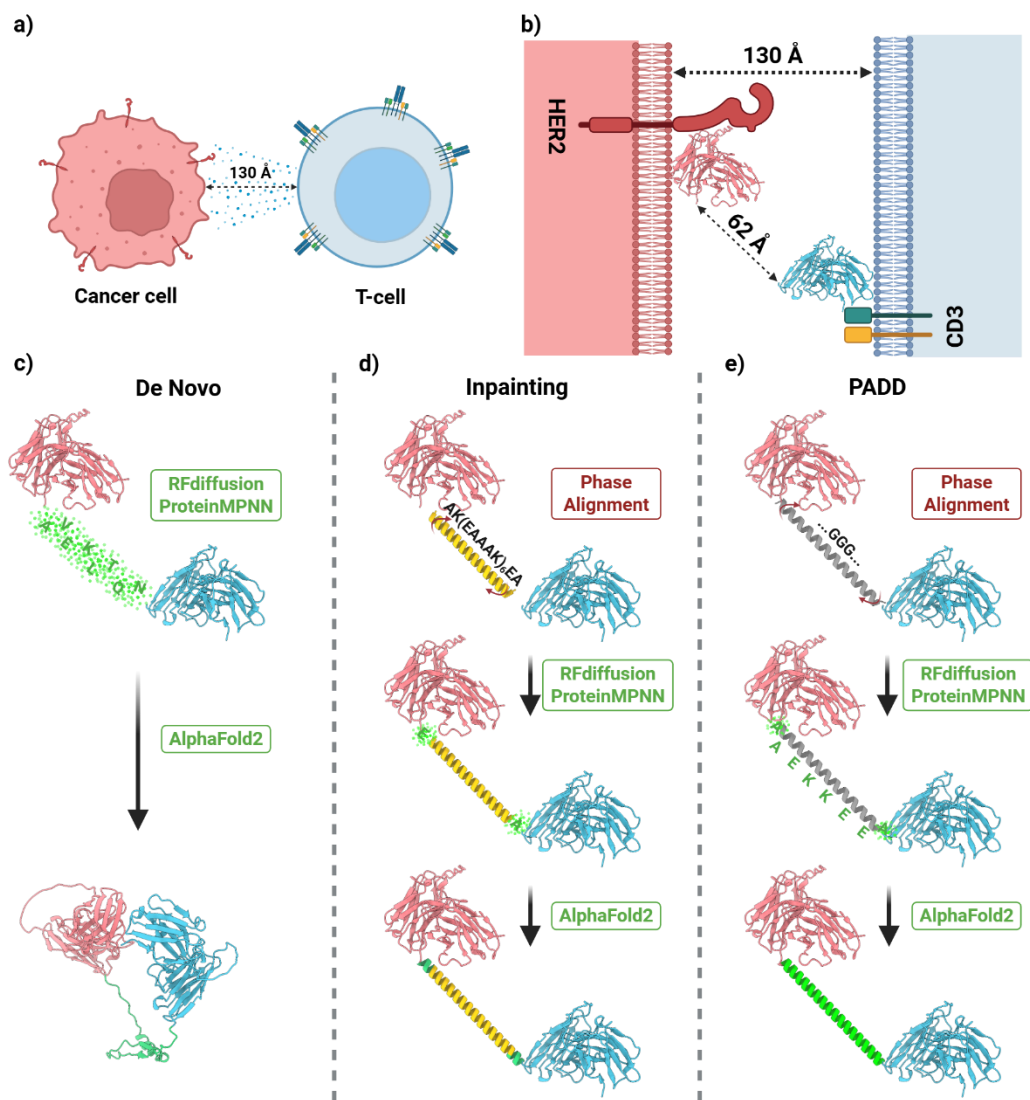


Figure 1. Design strategy and computational workflow. (a) Immunological synapse with optimal 130 Å inter-membrane distance for T-cell activation. (b) BiTE geometry requiring 62 Å linker span between scFv domains. (c) De novo generation: unconstrained RFdiffusion fails to achieve target geometry. (d) Inpainting: RFdiffusion solves junction geometry within human-defined constraints. (e) PADD: fixed phase-aligned backbone with ProteinMPNN sequence optimization.

2.3 Geometry-guided inpainting

The inpainting approach tested whether AI could solve local geometric problems when provided with a partially defined structural framework (Figure 1d). An initial template was constructed by positioning a theoretical (EAAAK)₈EA helical linker (42 residues, ~62 Å) between the two scFvs. The angular orientation was manually optimized using UCSF ChimeraX to minimize steric clashes while maintaining target distance. To create gap regions for AI-mediated problem-solving, 3 residues were deleted from the N-terminal junction and 5 from the C-terminal junction, leaving a 34-residue partial helix. RFdiffusion inpainting mode generated backbone coordinates with secondary structure bias toward rigid helices. ProteinMPNN sampled ~32 sequences per backbone at temperature 0.1, excluding Cys, Gly, and Pro to ensure linker rigidity. Candidates with RMSD < 20 Å proceeded to AlphaFold2 validation.

2.4 Phase-aligned de novo design (PADD)

PADD evaluated whether strict human-defined geometric constraints could yield superior rigidity when paired with AI-optimized sequences (Figure 1e). A Phase-Aligned Poly-Glycine Scaffold (42 residues) was constructed with mathematically calculated helical phase ($100^\circ/\text{residue}$), ensuring terminal vectors perfectly aligned with scFv attachment points to minimize torsional strain. RFdiffusion refined local geometry while maintaining fixed backbone topology. ProteinMPNN sampled sequences at increased temperature (0.3) to encourage broader exploration of sequence space. Candidates with $\text{RMSD} < 30 \text{ \AA}$ proceeded to validation.

2.5 Computational validation pipeline

All candidates passing initial filtration underwent AlphaFold2 structure prediction via ColabFold (Fast MSA mode, 3 recycles). RMSD was calculated using a two-way alignment protocol: structures were aligned on scFv1 and RMSD measured for scFv2, then vice versa, with the minimum value reported to account for potential domain flexibility. Confidence metrics included global pLDDT, linker-specific pLDDT (residues 241-282), and junction pLDDT (5 residues flanking each scFv-linker interface). Inter-domain PAE was computed for pairwise relationships between scFv1 and scFv2 residues to assess confidence in relative domain orientation.

Dynamic rigidity was quantified via NMA using ProDy (Gaussian Network Model, $10 \text{ \AA}$ cutoff). Effective stiffness was calculated from the eigenvalue spectrum of the Kirchhoff matrix, with higher values indicating greater resistance to deformation. The key metric "inter-domain fluctuation" was defined as the standard deviation of the distance between scFv1 and scFv2 centers of mass across the first 20 non-trivial normal modes, weighted by mode contribution; lower values indicate more rigid inter-domain coupling. Composite scores weighted: distance error (0.30), fluctuation (0.25), 1-pLDDT (0.20), PAE (0.15), RMSD (0.10). Top candidates underwent stress testing with extended AlphaFold2 predictions (12 recycles to test conformational convergence, 4 independent random seeds, 5 predictions per candidate).

3 Results

3.1 Failure of unconstrained AI design

The de novo approach produced uniformly poor results across all 160 structures spanning five linker lengths (50, 55, 60, 65, and 70 amino acids). The mean RMSD to the $130 \text{ \AA}$ Reference State was $51.4 \pm 8.1 \text{ \AA}$, with no structure achieving an RMSD below $36 \text{ \AA}$ (Figure 2a). This failure was consistent across all linker lengths, indicating that simply increasing linker length cannot compensate for the lack of geometric constraints. These results demonstrate that RFdiffusion, when given complete freedom, does not spontaneously generate the precise geometry required for BiTE function—validating the necessity of human-guided constraint specification.

3.2 Guided approaches show complementary strengths

Both guided approaches produced viable candidates. After quality filtering, inpainting yielded 68 unique structures ($\text{RMSD} < 20 \text{ \AA}$), while PADD yielded 106 structures ($\text{RMSD} < 30 \text{ \AA}$). Inpainting demonstrated clear advantage in achieving proximity to the $130 \text{ \AA}$ target (Figure 2b): mean distance error was $49.7 \pm 42.0 \text{ \AA}$ versus $96.3 \pm 41.2 \text{ \AA}$ for PADD ($p < 0.001$). This advantage was pronounced at stringent thresholds: 29 inpainting candidates achieved distance errors below $20 \text{ \AA}$, compared to only 1 PADD candidate.

However, PADD demonstrated superior structural confidence (Figures 2c, 2d). Mean linker pLDDT was significantly higher for PADD (73.4 ± 6.7) compared to inpainting (71.2 ± 2.2 ; $p = 0.014$). PADD also exhibited lower AlphaFold RMSD ($61.0 \pm 22.5 \text{ \AA}$ vs. $105.7 \pm 24.0 \text{ \AA}$; $p < 0.001$). Notably, 9 PADD candidates achieved the ideal combination of distance error $< 50 \text{ \AA}$ and pLDDT > 75 , while no inpainting candidates met both criteria. Inter-domain fluctuation showed no significant difference ($14.2 \pm 5.4 \text{ \AA}$ vs. $14.8 \pm 6.1 \text{ \AA}$, $p = 0.43$; Figure 2c). The accuracy-confidence trade-off is visualized in Figure 2e.

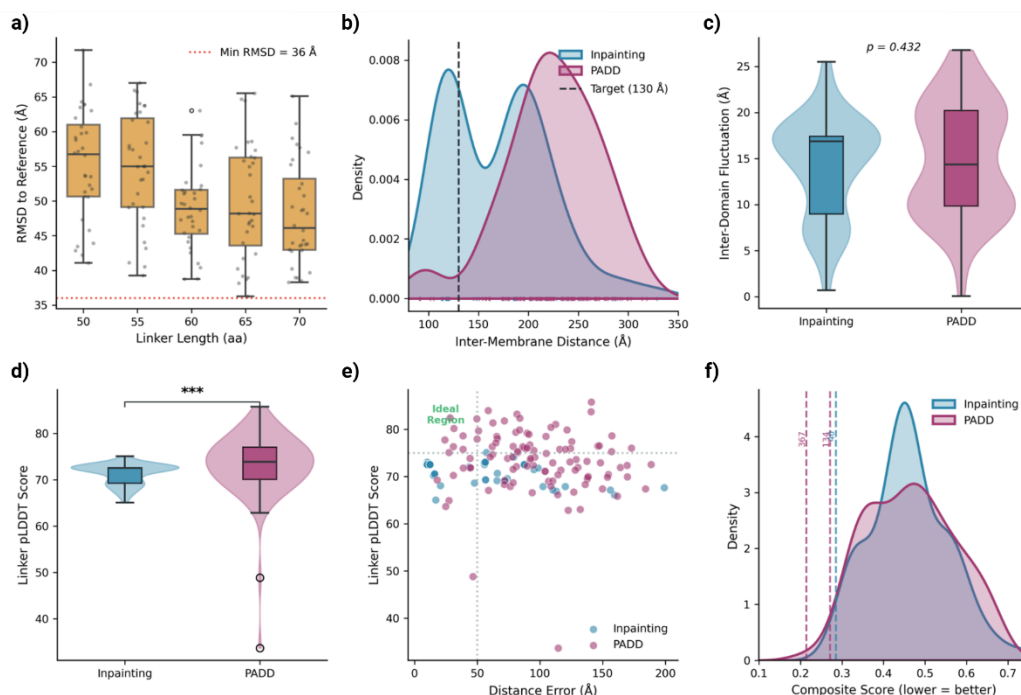


Figure 2. Comparative performance analysis. (a) De novo generation failure: RMSD across linker lengths, minimum 36 Å. (b) Distance distributions; inpainting clusters near 130 Å target. (c) Inter-domain fluctuation ($p = 0.43$). (d) Linker pLDDT; PADD significantly higher ($***p < 0.001$). (e) Accuracy-confidence trade-off; ideal region marked. (f) Composite score distributions with top candidates indicated.

3.3 Lead candidate identification and characterization

Despite different strengths, mean composite scores were nearly identical: 0.463 ± 0.095 for inpainting and 0.466 ± 0.106 for PADD ($p = 0.31$; Figure 2f). The top three candidates are presented in Table 1.

Table 1. Top 3 candidate profiles by composite score

Candidate	Method	Distance (Å)	Error (Å)	Fluct. (Å)	pLDDT	Score
PADD 85	PADD	179.7	49.7	8.00	82.2	0.214
PADD 54	PADD	101.7	28.3	12.41	82.4	0.271
Inpainting 5	Inpainting	113.2	16.8	1.67	65.1	0.285

Error = $|Distance - 130|$; Fluct. = inter-domain fluctuation; Score = composite (lower is better).

PADD 85 achieved the best composite score (0.214) through excellent rigidity (8.00 Å fluctuation), high pLDDT (82.2), and lowest PAE (26.43), despite distance overshoot (179.7 Å). **PADD 54** achieved closest PADD distance to target (101.7 Å) with maintained confidence. **Inpainting 5** achieved the lowest inter-domain fluctuation among all 174 candidates (1.67 Å) and closest proximity to target (113.2 Å, error = 16.8 Å), but with lower pLDDT (65.1). The bent junction geometry discovered by inpainting satisfies geometric constraints exceptionally well but represents a conformation that AlphaFold2 is uncertain about.

3.4 Extended validation reveals inpainting superiority

Stress testing (12 recycles, 4 seeds, 5 predictions) revealed a critical finding (Table 2). **Inpainting 5 demonstrated markedly superior reproducibility**: mean distance 130.8 Å (virtually identical to target) with SD = 29.5 Å (CV = 22.6%), and 4/5 predictions within ± 50 Å of target (range: 115.0–183.4 Å).

In contrast, PADD candidates exhibited extreme variability. PADD 85 showed mean distance 215.4 Å with SD = 69.5 Å (CV = 32.2%), ranging from 115.6 to 283.8 Å—only 1/5 predictions within acceptable range. PADD 54 performed similarly (mean 229.6 Å, SD = 62.8 Å, 2/5 within range). All candidates maintained high effective stiffness (>0.93), confirming linker rigidity per se was not compromised.

Table 2. Extended validation results (12 recycles, 5 predictions)

Candidate	Method	Distance (Å)	SD (Å)	Range (Å)	Within ± 50 Å
PADD 85	PADD	215.4	69.5	115.6–283.8	1/5
PADD 54	PADD	229.6	62.8	156.6–283.0	2/5
Inpainting 5	Inpainting	130.8	29.5	115.0–183.4	4/5

Bold indicates best performance. SD = standard deviation; Within ± 50 Å = predictions with distance 80–180 Å.

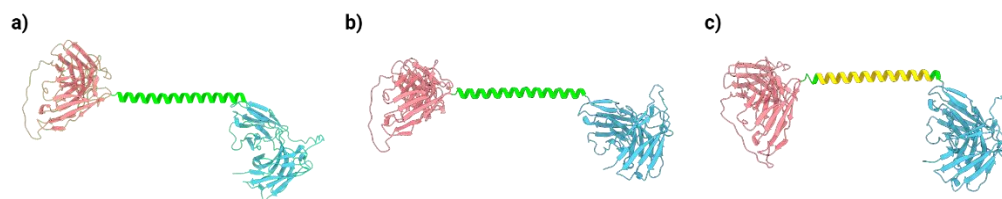


Figure 3. AlphaFold2-predicted structures of top 3 candidates (relaxed, rank 1). (a) PADD 85: linear helix with high confidence but distance overshoot. (b) PADD 54: closest PADD distance to target. (c) Inpainting 5: bent junction geometry achieving optimal distance with exceptional rigidity. Anti-CD3 scFv (pink), anti-HER2 scFv (cyan), designed linker (green/yellow).

4 Discussion

4.1 Human geometric constraints are essential

The complete failure of unconstrained de novo generation (mean RMSD > 50 Å across 160 structures) demonstrates a fundamental limitation of current AI protein design tools. RFdiffusion optimizes for general protein stability metrics learned from natural protein structures, but the BiTE design problem requires satisfying specific spatial constraints that are not represented in typical training data. This finding has important implications: for constrained design problems where specific spatial requirements must be met, human domain knowledge remains essential for defining geometric parameters that AI cannot infer from training data alone.

4.2 Reinterpreting the accuracy-confidence trade-off

Initial screening revealed what appeared to be a fundamental trade-off between geometric precision (inpainting) and prediction confidence (PADD). However, extended validation (Table 2) demands critical reinterpretation. The high pLDDT scores of PADD candidates may reflect AlphaFold2's familiarity with canonical helical structures rather than genuine stability. Under stress testing, PADD candidates exhibited extreme conformational variability (SD = 62–70 Å), suggesting the enforced linear helix does not represent a robust

energy minimum in the full BiTE context.

In contrast, Inpainting 5 demonstrated remarkable consistency, converging near the 130 Å target (mean = 130.8 Å, SD = 29.5 Å). The bent junction geometry, despite lower initial confidence, represents a more stable structural solution. We propose that lower pLDDT reflects AlphaFold2's unfamiliarity with engineered non-natural conformations—not genuine instability. This interpretation has significant implications for how the field should use AlphaFold2 confidence metrics when evaluating designed proteins that deliberately deviate from natural folds.

4.3 Mechanistic interpretation of structural solutions

The contrasting structural solutions provide mechanistic insight. Inpainting allowed RFdiffusion to introduce bends at scFv-linker junctions, effectively shortening end-to-end distance while maintaining residue count. These non-canonical conformations satisfy distance constraints but are less familiar to AlphaFold2. PADD enforced an ideal linear α -helix that AlphaFold2 recognizes as canonical, yielding higher confidence, but the linear geometry proves incompatible with the 130 Å target given fixed scFv orientations. The high variability under extended prediction suggests AlphaFold2 attempts to resolve this geometric incompatibility by sampling alternative domain arrangements.

4.4 Implications for AI-assisted protein design

Our findings have broader implications for AI-assisted therapeutic protein design. The failure of unconstrained generation, combined with the superior performance of appropriately constrained inpainting, suggests that the optimal human-AI partnership involves strategic task allocation: humans should define the critical geometric and functional constraints derived from mechanistic understanding, while AI should be granted freedom to explore solutions within those constraints. This is fundamentally different from either fully autonomous AI design or rigid human specification of every structural detail.

Based on these results, we propose a tiered strategy for AI-assisted therapeutic protein design: (1) use human domain knowledge to define critical geometric constraints based on mechanistic understanding of the therapeutic target; (2) allow AI limited freedom to optimize local structural features, particularly at challenging junction regions where rule-based design struggles; and (3) validate extensively with multiple prediction conditions and stress testing rather than relying solely on single-condition confidence metrics, which may not reflect genuine stability for non-natural structures.

4.5 Limitations and future directions

Several limitations should be acknowledged. All validation is computational; experimental expression, purification, and functional testing are required. AlphaFold2 metrics may not accurately reflect foldability of non-natural structures. The generalizability to other BiTE targets or linker architectures remains to be established. Future work should explore hybrid approaches combining inpainting's geometric precision with strategies to improve confidence, molecular dynamics validation, and experimental characterization of top candidates including biophysical measurements and functional assays.

5 Conclusions

We systematically compared three approaches for rigid BiTE linker design. De novo generation failed completely, confirming human constraints are essential. Initial screening suggested a trade-off between geometric precision and prediction confidence, but extended validation revealed inpainting produces more reproducible structures achieving the 130 Å target precisely. The apparent confidence advantage of PADD did not translate to stability under rigorous testing. We recommend Inpainting 5 as the lead candidate for experimental validation, representing optimal balance of geometric precision, structural rigidity, and prediction reproducibility.

Code availability

All scripts are available at <https://github.com/BiTE-Design>

References

- [1] Zhou X, et al. Front Immunol. 2024;15:1323049.
- [2] Baeuerle PA, Wesche H. Curr Opin Oncol. 2022;34:552-558.

- [3] Leithner A, et al. PNAS. 2025;122:e2420864122.
- [4] Nakamura K, et al. Nat Biotechnol. 2025;43:89-98.
- [5] Guidi M, et al. Nat Rev Clin Oncol. 2025;22:45-63.
- [6] Dewaele M, Fernandes RA. Nat Rev Drug Discov. 2025;24:127-145.
- [7] Chen W, et al. mAbs. 2021;13:1881449.
- [8] Radtke T, et al. Clin Pharmacol Ther. 2024;115:345-358.
- [9] Klein JS, et al. Protein Eng Des Sel. 2014;27:325-330.
- [10] Kjaergaard M. Biochemistry. 2021;60:1743-1751.
- [11] Patel A, et al. Biotechnol Bioeng. 2020;117:3923-3942.
- [12] Santich BH, et al. Sci Transl Med. 2020;12:eaax1315.
- [13] Arai R, et al. Protein Eng Des Sel. 2001;14:529-532.
- [14] George RA, Heringa J. Protein Eng. 2002;15:871-879.
- [15] Zhang Y, et al. ACS Chem Biol. 2024;19:916-925.
- [16] Park S, et al. Nat Commun. 2024;15:2847.
- [17] Voynov V, et al. mAbs. 2020;12:1816302.
- [18] Watson JL, et al. Nature. 2023;620:1089-1100.
- [19] Yim J, et al. ICML. 2023;40:39901-39915.
- [20] Zheng L, et al. Nat Mach Intell. 2024;6:287-299.
- [21] Dauparas J, et al. Science. 2022;378:49-56.
- [22] Gao Z, et al. arXiv. 2023;arXiv:2301.09642.
- [23] Jumper J, et al. Nature. 2021;596:583-589.

AI Co-Scientist Challenge Korea Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for NA).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will be asked to also include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While "[Yes]" is generally preferable to "[No]", it is perfectly acceptable to answer "[No]" provided a proper justification is given (e.g., "error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering "[No]" or "[N/A]" is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading “AI Co-Scientist Challenge Korea paper checklist”,**
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers.**

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The main claims
regarding the geometric
superiority of AI-guided
inpainting and the trade-offs
between precision and
confidence are stated in the
Abstract and Section 1.

Guidelines:

- The answer NA means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how

much the results can be expected to generalize to other settings.

- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: We explicitly discuss the lack of experimental wet-lab validation as a primary limitation in Section 4 (Discussion).

Guidelines:

- The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [NA]

Justification: This study focuses on empirical computational design and analysis, not on theoretical proofs.

Guidelines:

- The answer NA means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented

by formal proofs provided in appendix or supplemental material.

- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [\[Yes\]](#)

Justification: All Python scripts for analysis and the generated datasets are available in the supplementary material and code repository.

Guidelines:

- The answer NA means that the paper does not include experiments.

- If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While AI Co-Scientist Challenge Korea does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: We provide all
Python analysis scripts, the
complete dataset of generated
linker designs (including PDB
structures and metric logs), and
detailed parameters for the
generative models in the
supplementary material.

Guidelines:

- The answer NA means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so “No” is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to

reproduce the results. See the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.

- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental Setting/Details

Question: Does the paper specify all the training and test details (e.g., data splits, hyper-parameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [\[Yes\]](#)

Justification: We specify all key parameters (e.g., number of recycles=12, seeds=4) for AlphaFold2 and RFdiffusion settings in Section 2 (Materials and Methods).

Guidelines:

- The answer NA means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [\[Yes\]](#)

Justification: We report mean values with standard deviations (SD) and use independent t-tests (or Mann-Whitney U tests) to calculate p-values, as detailed in Section 3.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).

- If error bars are reported in tables or plots, The authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [\[Yes\]](#)

Justification: We utilized cloud-based GPU resources (e.g., Google Colab with NVIDIA A100/V100) for AlphaFold2 predictions, as noted in Section 2.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.

- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code Of Ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://nips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: We adhere to the NeurIPS Code of Ethics; our work aims to advance therapeutic antibody design for public health benefits.

Guidelines:

- The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader Impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: We discuss the potential positive impact on cancer immunotherapy efficacy in the Introduction and Conclusion sections.

Guidelines:

- The answer NA means that there is no societal impact of the work performed.
- If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from

feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [NA]

Justification: The released linker designs do not pose high risks for misuse (e.g., bioweapons), so specific safeguards are not required.

Guidelines:

- The answer NA means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [\[Yes\]](#)

Justification: We properly cite and respect the licenses of utilized tools (AlphaFold2, RFDiffusion) and public databases (RCSB PDB).

Guidelines:

- The answer NA means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New Assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [\[Yes\]](#)

Justification: New assets (generated linker structures and analysis scripts) are documented and provided in the supplementary material.

Guidelines:

- The answer NA means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.

- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and Research with Human Subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [NA]

Justification: This study
involves computational protein
design and does not involve
crowdsourcing or human
subjects.

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer:[NA]

Justification: This study does not involve human subjects, so IRB approval is not applicable.

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.